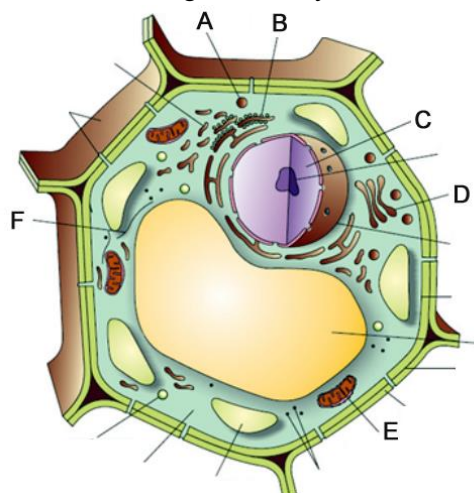


8. CELL: THE UNIT OF LIFE

1. Chief advantage of multicellularity over unicellular organization is
 - a. Attainment of larger size
 - b. Ability to succeed in struggle for existence
 - c. Better adaptation in relation to environment
 - d. Division of labour
 - e. All the above
2. The scientist who first observed and described a live cell is
 - a. Robert Hooke
 - b. Leeuwenhoek
 - c. Schleiden
 - d. Schwann
3. Cell theory was formulated by
 - a. Matthias Schleiden & Theodore Schwann
 - b. Rudolf Virchow
 - c. Anton van Leeuwenhoek
 - d. Robert Brown
4. Which of the following statements is incorrect?
 - a. Schleiden observed that all plants are composed of various types of cells.
 - b. Schwann proposed that animals and plants are composed of cells and cell products.
 - c. Schleiden found that plant cells have cell wall.
 - d. Rudolf Virchow proposed the concept of *Omnis cellula-e cellula*.
5. All cells contain
 - a. Cytoplasm and mitochondria
 - b. Lysosomes and ribosomes
 - c. Cytoplasm and ribosomes
 - d. Nucleus and endoplasmic reticulum
6. Size of bacteria may be
 - a. 30 to 50 μm
 - b. 300 to 500 μm
 - c. 0.3 to 0.5 μm
 - d. 3 to 5 μm
7. Consider the following statements about ribosomes
 - i. Ribosomes are non-membrane bound organelles.
 - ii. Ribosomes are seen in cytoplasm, chloroplasts, mitochondria and on RER.
 - iii. In prokaryotes, ribosomes are 80 S type.
 - iv. They are site of translation process.
 Of these,
 - a. All statements are correct
 - b. All statements except iii are correct
 - c. Statements i & ii only correct
 - d. All statements are incorrect
8. Match the following

A	B
1. Glycocalyx	A. Gas vacuoles
2. Mesosome	B. Slime or capsule
3. Chromatophores	C. Cell wall formation
4. Inclusion bodies	D. Pigments

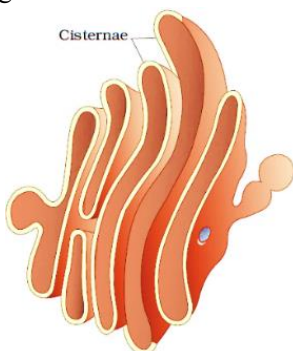
 - a. 1 2 3 4
 - a. B C D A
 - b. C D A B
 - c. A C B D
 - d. D A B C
9. Correct order of layers of cell envelope in prokaryotic cells from inner to outer is
 - a. Plasma membrane, cell wall, glycocalyx
 - b. Glycocalyx, cell wall, Plasma membrane
 - c. Plasma membrane, glycocalyx, cell wall
 - d. Cell wall, Plasma membrane, glycocalyx
10. Which of the following has no prokaryotic cell?
 - a. Bacteria
 - b. PPLO
 - c. Amoeba
 - d. Blue green algae
11. Analyze the following statements about extension structures in prokaryotic cells
 - i. Flagella are thin filamentous extensions having filament, hook and basal body from the cell wall.
 - ii. Hook is the longest portion of flagellum
 - iii. Pili and fimbriae have no role in motility
 - iv. Fimbriae are small bristle like fibres made up of pilin.
 Of these,
 - a. Statements i and iii are wrong
 - b. Statements i and ii are wrong
 - c. Statements i, ii and iv are wrong
 - d. Statements ii and iv are wrong
12. Polysomes are
 - a. Several lysosomes attached to a single mRNA to form a chain
 - b. Several ribosomes attached to a single tRNA to form a chain
 - c. Several ribosomes attached to a single mRNA to form a chain
 - d. Several lysosomes attached to a single tRNA to form a chain
13. Which of the following is correctly labelled?



- a. A = Ribosome, B= SER, C= Nucleolus, D= Golgi apparatus, E=Chloroplast, F= Microtubule
- b. A = Lysosome, B= SER, C= Nucleus, D= Golgi apparatus, E=Chloroplast, F= Centriole
- c. A = Microtubule, B= RER, C= Nucleolus, D= Golgi apparatus, E=Mitochondrion, F= lysosome
- d. A = Lysosome, B= RER, C= Nucleus, D= Golgi apparatus, E=Mitochondrion, F= Microtubule

14. Fluid Mosaic Model of cell membrane is proposed by
 a. George Palade b. Singer & Nicolson
 c. Robert Brown d. Camillo Golgi
15. In human RBC, the cell membrane has
 a. 52% protein and 40% lipids
 b. 40% protein and 52% lipids
 c. 32% protein and 65% lipids
 d. 65% protein and 32% lipids
16. A: Polar molecules require membrane carrier protein for transport.
 B: Polar molecules alone cannot pass through the non-polar lipid bilayer.
 a. A and B are correct and B is the correct explanation for A.
 b. A and B are correct and B is not correct explanation for A.
 c. A is correct but B is wrong.
 d. Both A and B are wrong.
17. Given below are some statements about cell wall. Select the wrong statement.
 a. It is a living rigid structure found outer to the plasma membrane of fungi and plants
 b. Cell wall of Algae is made of cellulose, galactans, mannans and minerals like CaCO_3 .
 c. In other plants, cell wall consists of cellulose, hemicellulose, pectins and proteins.
 d. Cell wall of a young plant cell (primary wall) is capable of growth.
18. Endomembrane system includes
 a. Endoplasmic reticulum (ER), Mitochondria, cell wall and vacuoles.
 b. Endoplasmic reticulum (ER), Golgi complex, lysosomes and vacuoles.
 c. Endoplasmic reticulum (ER), Golgi complex, Ribosomes and cell wall.
 d. Endoplasmic reticulum (ER), Mitochondria, lysosomes and cell wall.
19. Rough ER is
 a. ER which bears lysosomes on the surface and is involved in protein synthesis.
 b. ER which does not bear lysosomes on the surface and is involved in lipid synthesis.
 c. ER which bears ribosomes on the surface and is involved in protein synthesis.
 d. ER which does not bear ribosomes on the surface and is involved in lipid synthesis.

20. The given figure is



- a. Mitochondria b. Golgi apparatus
 c. Chloroplast d. Endoplasmic reticulum
21. A: Golgi apparatus remains in close association with the endoplasmic reticulum.
 B: Materials to be packaged as vesicles from the ER fuse with the *cis* face and move towards the *trans* face of Golgi apparatus.
 a. A and B are correct and B is the correct explanation for A.
 b. A and B are correct and B is not correct explanation for A.
 c. A is correct but B is wrong.
 d. Both A and B are wrong.
22. The membrane bound vesicular structures formed by the process of packaging in the Golgi apparatus is
 a. Mitochondria b. Vacuoles
 c. Lysosomes d. Plastids
23. Select the false statement in relation to lysosomes.
 a. These are membrane bound vesicular structures
 b. Lysosomes are called suicide bags of the cell.
 c. Lysosomal vesicles contain hydrolytic enzymes such as lipases, proteases and carbohydrases.
 d. Lysosomal enzymes are active at alkaline pH.
24. Match the following

A	B
1. Power house of cell	A. Vacuole
2. Suicidal bag	B. Chloroplast
3. Tonoplast	C. Mitochondria
4. Thylakoids	D. Lysosome

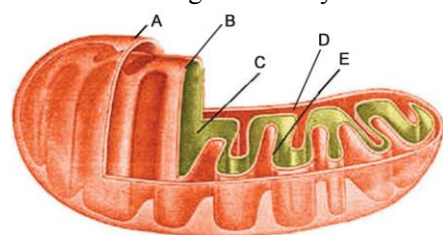
- | | | | | |
|----|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| a. | B | C | D | A |
| b. | C | D | A | B |
| c. | A | C | B | D |
| d. | D | A | B | C |

25. Match the following

A	B
1. Amyloplasts	A. Oils and fats
2. Elaioplasts	B. Proteins
3. Aleuroplasts	C. Pigments
4. Chromoplasts	D. Starch

- | | | | | |
|----|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| a. | B | C | D | A |
| b. | C | D | A | B |
| c. | A | C | B | D |
| d. | D | A | B | C |

26. Which of the following is correctly labelled?



- a. A= Outer membrane, B= Crista, C= Matrix, D= Inter-membrane space, E= Inner membrane

- b. A= Outer membrane, B= Inner membrane, C= Matrix, D= Inter-membrane space, E= Crista
- c. A= Outer membrane, B= Inter-membrane space, C= Matrix, D= Inner membrane E= Crista
- d. A= Outer membrane, B= Inner membrane, C= Crista, D= Matrix, E= Inter-membrane space
27. Matrix of mitochondria possesses
- A circular DNA, a few RNA molecules, 70S ribosomes & components for protein synthesis.
 - A circular DNA, a few RNA molecules, 80S ribosomes & components for lipid synthesis.
 - A linear DNA, a few RNA molecules, 70S ribosomes & components for lipid synthesis.
 - A linear DNA, a few RNA molecules, 80S ribosomes & components for protein synthesis.
28. Select the odd one
- Thylakoids
 - Chloroplasts
 - Crista
 - Grana
29. Which is true statement regarding size of ribosomes in chloroplasts and cytoplasm?
- Cytoplasmic ribosomes are smaller (70S) than the Ribosomes of the chloroplasts (80S).
 - Ribosomes of the chloroplasts are smaller (70S) than the cytoplasmic ribosomes (80S).
 - Ribosomes of the chloroplasts and cytoplasmic ribosomes are equal sized and 80S type.
 - Ribosomes of the chloroplasts and cytoplasmic ribosomes are equal sized and 70S type.
30. In chloroplasts, Chlorophyll pigments are present in the
- Thylakoids
 - Inner membrane
 - Stroma
 - Stroma and Grana
31. Select the mismatching pair
- George Palade – Chromatin
 - Cytoskeleton – Mechanical support
 - Axoneme – 9+2 array
 - Nucleolus – rRNA synthesis
32. To maintain surface area to volume ratio in a balanced state, some cells have acquired additional structures in the form of projections called
- Microtubules
 - Microfilaments
 - Microvilli
 - Lamellation
33. Eukaryotic ribosome is
- 60S + 40S = 80S
 - 50S + 30S = 70S
 - 60S + 40S = 90S
 - 50S + 30S = 80S
34. Pick out the false statement
- In animal cells, plastids are absent
 - In plant cells, centrioles are present
 - In animal cells, small vacuoles are present
 - In plant cells, lysosomes are rare
35. Which one of the following sets of cells are lacking nucleus?
- Mammalian RBC, fibroblasts and sieve tube cells of vascular plants
 - Mammalian RBC, platelets and sclerenchyma cells of vascular plants
 - Mammalian RBC, platelets and sieve tube cells of vascular plants
 - Mammalian RBC, macrophages & parenchyma cells of vascular plants
36. Chromatin contains
- DNA, histones, lipids & carbohydrates
 - DNA, RNA, carbohydrates & histones
 - DNA, non-histone proteins & RNA only
 - DNA, histones, non-histone proteins & RNA
37. Primary constriction of chromosomes is called
- Centriole
 - Centromere
 - Nucleosome
 - Satellite
38. On the sides of centromere there are disc shaped structures known as
- Kinetochores
 - Chromatins
 - Chromatids
 - Telomeres
39. Match the following
- | A | B |
|-------------------------------|------------------------------------|
| 1. Metacentric chromosome | A. Centromere is nearer to one end |
| 2. Sub-metacentric chromosome | B. Centromere is close to its end |
| 3. Acrocentric chromosome | C. Terminal centromere |
| 4. Telocentric chromosome | D. Middle centromere |
40. The main arena of cellular activities in both the plant and animal cells is
- Nucleus
 - Nucleolus
 - Cytoplasm
 - Endoplasmic reticulum
41. Regarding prokaryotes
- Generally smaller and multiply more rapidly than the eukaryotic cells
 - Prokaryotes have a cell wall surrounding the cell membrane
 - The fluid matrix filling the cell is the cytoplasm and there is no well-defined nucleus
 - The genetic material is basically naked, not enveloped by a nuclear membrane.
 - Many bacteria have small circular DNA outside the genomic DNA, called plasmid
 - Mesosome is the essentially infoldings of the cell membrane.
 - Only 3 & 5 are correct
 - Only 1,4 & 5 are correct
 - Only 3 & 5 are correct
 - Only 1, 2 & 3 are correct
 - All are correct

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